

Compose capabilities. Observe the product. Grade the evidence.

Capability-first federation, machine interfaces and user-path bridges, fine visual verification, canonical design state, full artifact traceability, and a defensible numeric grade.

Planning and reference implementation | September 16, 2026

This amendment preserves the existing owner allocation and technical program. It does not implement the native game bridge, generate approved UI mockups, run product pilots, certify a live product grade, or grant repository/deployment permissions.

One chat owns each current subject; independent evidence and review remain required. The original v1.3 archive is unchanged. Use the complete docs/ tree and per-owner PROOF-AND-GRADE.md as the actionable source.

The main correction

A shared capability can become part of a host without displaying another application. The useful proof is the real user outcome. A numeric score summarizes a declared, versioned assignment; critical failures still block acceptance.

Reading map

1 Capability composition · 2 Agent bridge · 3 Fine visual proof · 4 Canonical designs · 5 Traceability · 6 Numerical grading · 7 Work/owners · 8 Primary sources

Capability-first composition: one product experience, several implementations

The corrected abstraction

Two apps may contribute to the same user outcome without showing two apps, any embedded window, a plugin panel or a visible applet. A host can acquire a new command, transform, search source, persistence operation, runtime, renderer, evaluator or workflow step. Its own interface expresses the result. An embedded surface is one implementation option, not the definition of integration.

A Tracera capability can create or inspect real governed work through AgilePlus while Tracera renders the interaction in its own language. BytePort can use ShareCLI supervision and show deployment health without a second supervisor UI. Melosviz can acquire a qualified render/export provider, leaving the user in one editing workflow. A game can use a shared terrain or replay capability without exposing the library's brand. These are proposed examples, not implemented-product claims.

The proof is behavior with the provider present, absent, replaced, revoked and failed. A screenshot of a combined shell does not prove any of it. Conversely, a valid composition with zero UI mounts is not less federated.

Composition planes

Plane	Unit and obligation
Build/source	Use an owned or external package through a published contract; no casual source copy.
In-process	Direct calls, shared types and appropriate ownership for low overhead.
Runtime service	Scoped local IPC or network requests where lifecycle/isolation justifies them.
Data/document	Operate on authoritative objects using stable IDs and explicit read/write contracts.
Workflow	Compose results, cancellation, approvals, compensation and progress across capabilities.
Interaction	Host-rendered commands, tools, modes and semantics; accessible user behavior.
Surface	Optional native view, isolated web surface, remote pixels or bounded applet panel.
Deployment	Discover installed providers, reconcile versions/scopes and activate qualified bindings.

These are not a maturity ladder. Use the least costly valid combination. Not every capability needs hot loading, a process, a registry, a universal database or network RPC. Optional dynamic resolution complements compile-time composition; it does not replace sound typed boundaries.

Two-way engineering responsibility

Every substantial project change examines both directions: what existing capabilities should this implementation consume, and what useful capability can it expose for others? Search actual owned packages and accepted consumers, external libraries, upstream forks and relevant research. Prefer native facilities and mature implementations; then adaptation, narrow wrapping, upstream patching and small maintained deltas. New custom logic is justified by a recorded semantic or operational benefit, not a preference to own every line.

Least custom code is not fewest source lines at all costs. A short opaque expression, a slow generic layer or a dependency with an incompatible security model can raise total burden. Preserve accepted performance, numerical behavior, failure semantics, compatibility, privacy, resource and accessibility properties. Compare ecosystem-wide maintenance, not just local LOC. Avoid universal wrappers that

duplicate a dependency's entire API.

Current supported consumers need real compatibility tests; committed future consumers need explicit contracts/spikes; plausible future consumers justify inexpensive seams, not speculative features. Unknown external consumers are not zero consumers. One owner chat is a write boundary, not a reasoning boundary. Coordinate shared package changes and migrations with their owners.

Hexagonal and recursive constraints

Pure domain behavior stays independent from concrete UIs, engine bindings, providers and deployment. Ports express semantic operations, not incidental vendor APIs. Adapters explicitly translate error, retry, cancellation and data behavior. Architecture fitness checks validate dependency direction and prohibited imports. Consumer-driven contract tests validate adapters. Fake adapters support narrow tests; real adapters must qualify integration and installed E2E.

A composite can expose a reduced interface and participate at the next level. Recursive identity does not imply recursive full-shell nesting or unrestricted permission inheritance. Distinguish installation graphs, service dependency graphs, data authority and user perspective. Discovery metadata is not consent; learned suggestions are not execution permission. Headless provider changes must remain diagnosable even when the integrated experience hides the component boundaries.

Required witness

First demonstrate a zero-mount capability contribution through an existing host command. Verify that the underlying real object or result changes, both sources agree, the host remains coherent, a revoked permission fails correctly, no optional peer breaks the standalone host, and uninstall/detach preserves authoritative data. Then try a second consumer so the port is not secretly one hardcoded pair. Do not build another mandatory App Center before this works.

Agent bridge: observation, action and proof are separate channels

Purpose and access classes

An agent needs to understand the product, act through its public experience, inspect actual outcomes and locate regressions. No single tool grants complete observability. The adapter advertises only capabilities it can perform and proves its own health, instance binding and behavior.

Class	Available leverage	Qualification rule
Owned/white box (Civis, owned GUI/CLI)	Domain API, engine instrumentation, controlled fixtures, state snapshots, trace/profiler hooks, accessibility and native capture	Internal checks supplement but never replace actual packaged user-path and rendered-output proof.
Gray box (Dino, WorldSphereMod and allowed mod/plugin paths)	Permitted mod SDK/hooks, bridge IPC, selected engine state, logs, public game files plus external input/capture	Reuse the existing game bridge; record base-game and mod identity and what remains inaccessible.
Black box (third-party application surface)	OS input, accessibility where exposed, screen/audio capture, process/log/file observers where authorized	Judge observable behavior; do not claim structural coverage of unavailable proprietary internals.

CivicSurvival/CivicWarfare has owned mod code and a separate host game. Classify that boundary explicitly instead of calling the entire host white box. Native desktop, browser, terminal and media applications use the same evidence contract with different adapters.

Three action channels

Fixture control creates a world, seeds data, positions an actor or stages a failure. It is useful test setup but cannot prove the UI action being tested.

Semantic product commands exercise the actual service/domain API and verify contracts efficiently. They must not be confused with a clicked button or physical input path.

User-path actions use the supported input route: OS keyboard/mouse/controller/PTY, or an explicitly declared accessibility/browser semantic action. Direct engine mutation is prohibited as a substitute for the user behavior being assessed. Clicking via an accessibility action is distinct from native pointer input; both can be required for different obligations.

A test may use all three, but records every switch. Setup changes are frozen before the target observation, except for deliberate fault/control operations. After input, observe dispatch, domain effect, rendered effect and persistence as applicable. 'Command accepted' is not 'operation completed'.

Session handshake and correlation

Bind every run to product and repository IDs; source and artifact digests; base game/mod/asset versions; supported build flags; process ID and start identity; window/render target; session nonce; selected world/save; principal/workspace; adapter version; graphics/audio profile; and actual permissions. Verify a unique harmless challenge/observation to avoid acting on a stale window or wrong named pipe.

Commands carry operation IDs, deadline, expected state/version where supported, authorization context, and explicit idempotency semantics. Events and captures carry sequence, monotonic time, simulation tick, render/presentation frame and origin. These clocks differ; record the mapping rather than equating an ECS update with a displayed frame. A fresh image hash alone cannot establish freshness when the scene is static.

Required semantics: capability discovery, observe, user action, semantic action, fixture control, wait-for predicate, frame/clip/audio acquisition, metrics, read-only diagnostic buffers, artifact export and cancellation. Names are conceptual contract operations, not claims of existing endpoints. No adapter reports success for an unsupported method.

Observation stack

- Semantic state: scene tree/ECS, objects, components, selection, resources, workflows and permissions where permitted.
- Actual output: final color frames or external window capture, video, audio, terminal output and accessibility structure.
- Diagnostics: logs, crashes, timing, draw-call/profile data and optional depth/normal/object-ID/motion buffers.
- Durability: saved data, migrated formats, restart/rejoin behavior, external side effects and produced artifacts.

Diagnostic buffers explain defects but do not replace final presentation. Internal state and rendered pixels can independently be wrong or stale. Cross-check them. For a visible feature, a log that its system initialized proves initialization only.

Isolation and long runs

Use per-run writable data, output paths, ports/pipes, caches and save copies; lease any native device or window. Hidden means private and nondisruptive, not permission bypass. Separate persistent authorized account/entitlement identity from disposable test state. Do not clone auth tokens or bypass DRM/anti-cheat/mutex protections to manufacture concurrency. Use supported launch/session mechanisms, independent entitled identities or serialize tests when necessary.

Do not capture the operator's desktop, private browser or live microphone as a convenient default. Use allowlisted windows/displays and redacted derivative artifacts with restricted raw masters. Bound CPU, GPU, RAM, disk and networking; keep gaming/Ableton workloads protected. Distinguish software-rendered CI evidence from qualification on the supported hardware profile.

Negative controls

Wrong window/pipe; stale nonce or capture; old binary; missing assets; no-op input; disabled input handler; unavailable bridge; scene transition destroying a hook; main-thread dispatch starvation; dropped acknowledgement; partial operation; process crash; save corruption; unauthorized action; cross-worker data access; unavailable GPU; headless logic without rendering; and injected known-bad visuals must produce honest failures or unresolved outcomes.

The bridge must fail before corrupting an unrelated instance. A lost acknowledgement triggers state reconciliation before retry. Never assume exactly-once external effects from transport delivery alone.

Reuse

Inspect DINOForge GameControlCli and existing named-pipe protocol, Civis capture/playthrough scripts, native accessibility facilities, Playwright, Bevy screenshot/render-target support, Godot viewport capture and Unreal/Unity testing facilities. Wrap qualifying adapters instead of writing a new per-repository transport. Put shared interfaces/normalizers in their accepted shared owner; game-specific hooks stay with the game/mod. Primary sources are listed in SOURCES.md. No native bridge was executed by this amendment.

Visual and perceptual verification: detecting small meaningful defects

A feature-sized contract

A full-frame similarity average can conceal a missing two-pixel seam, one-pixel icon, short flicker, invisible outline or wrong resource digit. Every high-risk visual capability needs a named observable witness: target object/region, camera/pose, scale, lighting/time, expected state, interval, render profile, mask policy, spatial/temporal tolerance, and meaningful unacceptable defects.

Track final frame pixels and, where available, semantic scene state and read-only diagnostic buffers. A visible target needs existence, position/visibility, appearance and temporal behavior tests, not just 'component is present'. A camera may deliberately expose the target during fixture setup; do not retarget it after a defect disappears from view to manufacture a passing shot.

Two capture profiles, both necessary

Controlled regression pins seed/checkpoint, resolution/DPI, fonts, content, camera, driver/engine/profile and bounded warm-up. It may remove approved noisy effects to isolate an algorithm or layout property.

Representative player profile retains the actual shipping anti-aliasing, animation, motion blur, LOD, lighting, dynamic resolution, audio and presentation path relevant to the promise. It tests the same feature under motion, overlap, extreme content and supported resource pressure. Turning off the feature to stabilize a screenshot cannot certify that feature.

Baseline compatibility is keyed to the complete render profile. A changed engine/driver or product design invalidates or segments a baseline; workers do not automatically accept a new screenshot when the old one fails. Empty baseline/capture sets produce NOT_ASSESSED or blocked acceptance, not success.

Complementary methods

Method	Best use	Limitation
Exact/pixel threshold	Stable text, icons and controlled renders	Brittle across uncontrolled rendering environments
ROI/edge/shape checks	Small seams, outlines, alignment, clipping, object silhouettes	Regions, masks and alignment require independent review
Perceptual similarity	Broader structural changes and noisy images	Global averages can hide tiny critical failures
Semantic/accessibility checks	Labels, focus, state, visibility semantics	A correct tree does not prove visible pixels
Depth/normal/object-ID buffers	Diagnose occlusion, geometry, material and picking issues	Internal buffer success does not prove final composition
Temporal windows	Flicker, TAA trails, LOD popping, animation stalls, response timing	Needs frame identity and sufficient sampling
Calibrated vision model	Interpreted layout/style/readability and unfamiliar defect triage	Fallible, prompt-sensitive and unsuitable as sole critical oracle
Audio signal/listening checks	Silence, clipping, sync, missing cues, glitches	File presence and decoded duration do not prove perceptual quality

Use bounded high-resolution crops in addition to full-context frames. Never enlarge a downsampled thumbnail and call it recovered pixel detail. Preserve original frames and lossless analysis derivatives; marketing video compression is a different output. Color checks must record transfer function, primaries, range, bit depth and tone mapping. An SDR screenshot cannot prove full HDR panel behavior.

Calibration and adversarial controls

Build a known-good/known-bad corpus per feature class. Seed a missing mesh, clipped label, wrong biome sample, disabled tooltip, tiny seam, flipped normal, hidden required icon, short flicker, stale frame, silent audio and nonfunctional input handler. Measure detector false acceptance and false rejection per class, not only a pooled score. Keep holdout cases separate from the development loop and retain all failed attempts.

For model-assisted judgments bind the model/version, rubric/prompt digest, actual image/clip payload digests, crop context and response. Reject fabricated screenshot paths, malformed output and confident narrative unsupported by captured media. Captured UI and logs can contain prompt injection; treat them as untrusted evidence. Do not allow a model's self-declared confidence to become a mathematical probability or a new permission.

Independent detectors can still have correlated blind spots. Disagreement is a finding, not an invitation to choose the most favorable result. Use analyst/operator sampling and adjudication for disputed perceptual claims; broad claims of fun, taste, accessibility or universal quality cannot be proven by one image model. The machine can produce useful repeatable proxy grades when their limits and calibration are explicit.

Product witnesses

Civis: placement ghost, selection, minimap indicator, terrain/water change, character animation, menu/HUD layering, camera transition and save/reopen continuity in the real game. Preserve subsystem reproducibility without imposing global determinism.

Dino: a distinctive real pack change, public user-path action, game-state effect and actual changed pixels; test main-menu/gameplay transitions, asset catalog failures and hook lifecycle.

WorldSphereMod: full camera orbit, seam/pole behavior, terrain coloring, LOD transitions, depth/collision/picking, animated actors and supported effects in the actual host game. A world coordinate may be correct while a projected surface is visibly wrong.

CivicWarfare: actual installed mod HUD, onboarding, combat/economy loop, visual feedback, save/reload, compatibility and clean disable under the supported host-game profile.

All other GUIs: design states, focus/navigation, accessibility, loading/empty/error/recovery paths and real content. CLI-only subjects use transcript/PTY assertions, machine-output contracts and VHS demonstrations rather than invented GUIs.

Canonical design state and executable mockups

Design is first-class state, not disposable chat output

Keep inspectable design decisions beside the product model, not solely in screenshots pasted into agent chats. The user should be able to inspect the current intended experience, alternatives and rejected options before and during implementation. An accepted design constrains work; it is not proof that the running implementation conforms.

Each design record links: human source and interpretation; user job/capability; required states and transitions; data fixtures; interface/interaction constraints; architecture assumptions; editable source and rendered previews; provenance/licensing; review state and exact version; implementation paths; acceptance scenarios; and actual resulting captures. Use subject-local source with reconciled Tracera entities rather than a hardcoded universal design database in a tool.

Representations by purpose

Representation	Appropriate use
ASCII/text sketch	Fast CLI layout, command transcript, flow and state discussion
SVG/Figma/Penpot/vector/raster	Editable screen composition, alternatives, typography and assets
Story/component fixture	Real interactive control behavior and state variants
Clickable prototype	Navigation and workflow validation before backend completion
Scene/blockout	Camera, spatial interaction, environment, character scale and lighting
Motion/audio specimen	Timing, transition, feedback, pacing and sound behavior
API/schema/state model	Nonvisual workflows, event ordering, data and error contracts
Runtime capture	Actual implementation evidence, never a substitute for the design's source

Choose the smallest useful fidelity; do not force Figma on an engine scene or a 3D prototype on a CLI. Reuse qualified shared tokens, components, generators and sources. Adobe/Blender/native tools are used where available and properly licensed. Font binaries and restricted third-party assets are not redistributed by default.

State coverage

For each applicable surface model default, empty, loading, partial, populated, success, error, denied, offline, reconnecting and recovery states. Include long/localized content, keyboard/IME, focus, accessibility, DPI/window sizes, themes and reduced motion where supported. Capture relationships rather than an unbounded Cartesian screenshot explosion; critical variants remain explicit.

For games, include build/inspect/command/possession modes, overlays, population scale, camera regimes and unattended-world return. A static landing-style mockup is not a gameplay interaction specification.

Lifecycle and creative freedom

Exploratory -> proposed -> reviewed -> accepted-for-scope -> implemented-candidate -> behavior-validated -> superseded. 'Reviewed' names actual feedback; 'accepted' needs scoped authority and version, not an agent changing a status file. Creative alternatives may coexist. Branch experiments are not defects merely because they differ from the accepted design.

User approval may be asynchronous. Agents can make reversible provisional proposals while waiting, but must not label them approved. Freeze only what is needed for the current slice; keep explicit design degrees of freedom so creative work is not reduced to pixel tracing.

Baselines and change control

Maintain a design baseline (intended outcome) and a runtime regression baseline (observed qualified artifact). A regression screenshot can preserve a bug; a mockup can portray an impossible or unfinished capability. Qualification checks both design intent and runtime contract.

Changing a failed baseline requires a legitimate design/compatibility decision, not just a new screenshot. Record impact on downstream consumers, shared tokens, documentation, journeys and grading scope. Changes to a shared component must re-evaluate affected hosts, not only the original app.

First slice

Choose one current user journey. Publish its editable flow plus fixture-backed states and proposed interactions. Bind the accepted version when approval exists. Implement one path, capture the actual result through the bridge, compare the result with relevant design and behavioral contracts, record dissatisfaction, and iterate. The design tools/records remain usable before the complete Tracera UI exists.

Typed traceability from human thought to adopted behavior

One evolving product model, multiple authoritative sources

Human words -> interpretation/assumption -> product job -> capability -> requirement/invariant -> design/architecture/contract -> source/config/asset -> test/evaluator -> actual run -> artifact -> installation/deployment -> observed user outcome -> dissatisfaction/change.

This is a graph, not a mandatory linear assembly line. Research can revise a design; a runtime incident can expose a missing requirement; a prototype can precede a finalized specification. Preserve that provenance and uncertainty instead of retroactively pretending every discovery was planned.

Subject repositories own declared source artifacts. Native engines own scoped mutable work state. Tools render, validate and reconcile records. Tracera owns its persistent product model and links; it does not secretly appropriate every underlying data authority. The inventory and grading views are projections, not separate competing registries.

Artifact families to account for

Exact prompts, intent synthesis, glossary/assumptions, persona/jobs, capability models, FR/NFR/system/interface requirements, domain/data/state models, HLD/ALD/LLD, ADRs, research/hypotheses/experiments, SOTA and pilot evidence, designs/mockups/fixtures/assets, source and generated code, dependencies/patches/licenses, contracts/configuration, schemas/migrations, tests/fuzz/property/mutation corpora, test tools/oracles, WBS/work/DAG/estimates, builds/packages/signatures/SBOMs, deployment/IaC/permissions/secrets references, runbooks/SLOs/restore procedures, docs/examples/translations, actual runs, performance/captures, releases/installations, incidents, feedback and accepted changes.

Account for meaningful semantic units and spans. Generated and vendored files inherit their producer/upstream contract and local delta. Do not handwrite a paragraph per token or require a thousand ADRs because there are a thousand scenarios. Expand real uncertainty and behavior, not arbitrary document counts.

Both directions are mandatory

Forward: every accepted requirement reaches appropriate implementation/design/tests/evidence or an honest unimplemented/blocked/research record.

Reverse: every public surface, significant source unit, dependency, asset, workflow and distributed artifact has a role, owner, rationale and applicable obligations. Find undocumented code as well as unimplemented requirements. A file moved under docs or excluded from a workspace is not thereby out of product scope.

Typed evidence prevents category errors

'Builds' proves buildability; it does not prove usability. 'System initialized' proves initialization; it does not prove visible content. 'Screenshot exists' proves capture existence; it does not prove its content, freshness or expected pixels. 'Mock passes' proves the mocked contract path; it does not prove the live provider. 'Approved mockup' proves intended design; it does not prove installation. 'All known FRs linked' proves closure against a known inventory; it cannot prove no requirement is missing.

Relations therefore carry subject, revision, profile, claim type, source locator, producer and timestamp. Work status, requirement status, evidence status, publication and adoption are independently represented. Historical evidence stays accessible but may no longer qualify a new candidate.

Specification adequacy and scope versioning

A grade needs an explicit rubric universe. Define artifact-role applicability, capability/state/failure/support matrices and risk boundaries before grading. Use independent gap analysis, user feedback and adversarial scenarios to find omissions. Record specification adequacy separately from implementation satisfaction. Do not certify completeness for all possible ideas.

Preserve full-horizon intent and a bounded current viable product scope. Both can have percentages, but always with their own scope/rubric IDs. Adding real requirements can lower the current percentage without negating useful progress. Show old-scope/new-scope and scope-change effects separately. Never erase requirements merely to improve the grade.

Refresh and interchange

Use stable IDs, content/version hashes and source anchors, not paths alone. Update affected subgraphs incrementally. A changed source or artifact marks dependent evidence stale unless an independently accepted applicability proof permits reuse. Cached local records do not establish current remote or installed state. Work engine receipts are queried from the actual scoped service rather than inferred from generated docs.

One owner can produce and maintain the graph through portable repo-local records now. The shared schema and adapters enable later ingestion without blocking real improvements on complete platform implementation. Recover SROC/CDP source definitions separately; this graph does not invent them.

An evidence-backed percentage and grade

Yes: calculate a literal grade

The program must provide a numerical result for a declared assignment, not only a qualitative narrative or traffic light. The assignment is a versioned product scope, rubric, support profile and candidate artifact. No score claims to measure every possible future feature or all subjective quality. Numerical grading is now explicitly required; it does not weaken critical gates or evidence honesty.

The following defaults are proposed program policy, not an accepted industry standard. Resolve them through the existing scorecard owner instead of launching a competing grading project. Preserve richer accepted rubric definitions and import them through adapters.

Freeze the denominator and weighting

Select applicable categories for the product role. A proposed GUI/game profile is: intent/design 12; specification/architecture 10; real behavior 20; independent assurance 18; user experience/rendering 14; safety/recovery/security 12; ecosystem reuse/composition 5; delivery/operations 5; documentation/research 4. Weights sum to 100. Infrastructure or CLI profiles differ explicitly; do not force a graphical denominator onto an API library.

Within category k , split its weight among declared criteria i . Let a_i be the normalized within-category weight, with $\sum a_i = 1$. Let x_i be the earned fraction from an admissible evaluator receipt, between 0 and 1. Missing, stale, wrong-scope, unexecuted or inadmissible evidence earns zero, while remaining visible as unknown/blocked rather than a demonstrated behavioral failure.

```
category_k = sum(a_i * x_i)
score = sum(category_weight_k * category_k)
```

The score is between 0 and 100. This is not the percentage of tests passing, percentage of lines visited, percentage of files documented or percentage of a real-world truth discovered. Each is a separate possible metric with its own denominator.

Proposed letter mapping: $A \geq 95$; $B \geq 85$; $C \geq 70$; $D \geq 50$; $F < 50$. A production profile may set stricter limits. Display the value and letter WITH the gate state: '96.2 / A - BLOCKED: save-corruption invariant failed'. Do not hide the blocker in a tooltip or cap the arithmetic invisibly. A grade never grants merge, release, deployment or delete authority.

Critical gates cannot be traded away

Every critical obligation needs full credit and all required modalities/action paths. Other mandatory checks satisfy their declared thresholds. Independent unit/integration/E2E 85% floors are separate mandatory cells; preserve higher accepted floors. High unit coverage does not compensate for low integration coverage. Documentation cannot compensate for a required launch, consent, persistence or isolation failure.

A missing required measurement makes acceptance UNASSESSED/UNVERIFIED; an established violated mandatory threshold makes it BLOCKED. Empty, not-run, skipped, errored or infrastructure-blocked checks never count as passes. A pass on a direct domain API cannot satisfy a required user-input interaction without that evidence channel.

Honest state beneath the number

Return at least: score and grade; gate state and failed/missing mandatory IDs; evidence-assessed weight; unknown/inadmissible weight; category vector; target/source/profile/rubric identifiers; timestamp; reference constraints and authenticated-authority status.

The optional 'known-score to all-unknown-perfect' range is an arithmetic bound over unassessed criteria, NOT a confidence interval or probability. A known failure stays a known failure in that range. Statistical intervals for stochastic benchmarks belong to their native metric reports. Do not multiply by the agent's claimed confidence.

Separately show current-CVP and full-horizon scope scores, design readiness, verified implementation, delivery readiness and operational evidence where required. They are projections over declared criteria, not an excuse to cherry-pick a higher headline. Product/library/CLI scores are comparable only under compatible rubric profiles and support boundaries.

A synthetic worked example

Assume the category vector (in the order above) is 0.95, 0.90, 0.85, 0.82, 0.70, 0.90, 0.80, 0.75, 0.90. With the proposed weights the score is 84.11 / C. If a mandatory save/load invariant fails, the status is BLOCKED regardless of the average. This is arithmetic illustration, not any repository's grade.

If newly discovered requirements enlarge the accepted scope, show scope revision N and N+1 and recompute a same-scope comparison where possible. Do not call a denominator change an implementation regression, and do not hide it to keep the score rising. Breakdowns should expose removed/added criteria, waived scope, invalidated evidence and actual behavior changes.

Prevent grading games

Fix category weights before measurement. Adding 500 easy tests cannot dilute a hard visual failure. Refactoring code into more files cannot raise purpose coverage merely by record multiplication. A criterion split conserves parent weight. Aliases and duplicate evidence are not new independent obligations. Critical status and N/A/exemption decisions are externally reviewed and versioned. Expired waivers reopen gates.

Do not accept arbitrary latest reports: bind candidate digest, source, support profile, oracle version and execution. A successful unit test of the reporter is not product evidence. Reject canned scores, screenshot paths without accessible media, test lists without execution, success-after-collection-error and automatically accepted baseline refreshes. Keep an adversarial holdout corpus and audit the grader independently from the implementer.

Qualitative and emergent properties

Some perceptual, creative and game-design goals use anchored model-assisted rubrics, user studies or human adjudication. These can produce useful repeatable grades when calibrated and labelled; they are not infallible machine proofs of fun, accessibility or taste. A machine-only proxy that is not qualified for a critical decision leaves that gate unresolved. Use actual media, specific judged criteria, multiple cases and explicit disagreement handling.

White-box products can supply deep structural measurements. Gray/black-box host-game internals are outside owned-source denominators under an approved profile, while the full observable user contract remains in scope. Do not pretend to measure inaccessible engine-line coverage or mark all graphical behavior N/A.

Reference implementation boundary

grade.py computes weighted scores and selected admission checks against local JSON receipts/artifact files. It verifies source/artifact/profile/oracle identity, age, permitted action modes, required modalities, counts, unique IDs and hashes. It cannot independently certify that a producer executed honestly, that a screenshot depicts the right scene, that a semantic judge is accurate, or that approval references are authentic.

All bundled examples use a synthetic subject. Output always states `product_qualified=false`, `producer_authenticity_verified=false`, and `release_authorized=false`. Integrate trusted runner attestation, independent review, calibration and the existing real rubric/engine before treating production scores as accepted. The grader is a small reference adapter, not a new SSOT service or a replacement for native validators.

Existing owners; small proof slices

Keep one owner chat per current non-zz subject. Refresh actual role, identity and lease; do not reactivate obsolete ten-seat/pair instructions. Independent verification remains required, but writing an approval string does not establish approval.

The following eight work packages are proposed and unestimated until the actual subject, source paths, environment and authority are bound. They do not authorize cross-repository writes, publication or destructive changes. Full requirements, negative controls, rollback and permission prerequisites are in proof/work-packages .json and specs/013 through specs/016.

WP-PROOF-13-1 — Capability-first composition: contract and first positive path

Bind one host command to a real owned or external provider through the least-cost valid boundary, with no peer-UI mount. Record provider/consumer contracts, shared object identity and before/after behavior.

Decisive counterexample: A valid in-process provider is not forced through an unnecessary network layer.

Dependency: No universal prerequisite; local scope/permissions still required.

WP-PROOF-13-2 — Capability-first composition: negative controls and qualified consumer

Run the same contract with the provider absent, failed, revoked and replaced; add a second compatible consumer and prove one authoritative result with no duplicate mutation or data copy.

Decisive counterexample: Mocked provider success cannot satisfy a required live integration.

Dependency: WP-PROOF-13-1.

WP-PROOF-14-1 — Agent bridge and graphical proof: contract and first positive path

Adapt one existing white-box or permitted gray-box bridge. Bind the actual candidate, process, window and nonce; correlate user action, application effect and final-frame capture without fixture shortcuts.

Decisive counterexample: Cross-worker capture/data access or missing license cannot be bypassed to pass.

Dependency: No universal prerequisite; local scope/permissions still required.

WP-PROOF-14-2 — Agent bridge and graphical proof: negative controls and qualified consumer

Exercise one owned-game witness and one permitted host-game witness with known-bad input/render controls, fine ROIs and temporal samples. Demonstrate stale-window/frame rejection and isolate foreground resources.

Decisive counterexample: Known-bad subtle examples must be caught at the predeclared required rates.

Dependency: WP-PROOF-14-1.

WP-PROOF-15-1 — Canonical design and complete artifact traceability: contract and first positive path

Canonicalize one editable UI/flow or scene design with populated/empty/error/recovery states and fixture data. Keep proposals and approval evidence separate; link the actual implementation and capture.

Decisive counterexample: Autoupdating a failed screenshot baseline without a decision cannot close the defect.

Dependency: No universal prerequisite; local scope/permissions still required.

WP-PROOF-15-2 — Canonical design and complete artifact traceability: negative controls and qualified consumer

Run forward and reverse gap checks across intent, contracts, assets, code, configuration, tests, runs and installed artifact. Change one design/source element and prove affected evidence becomes stale without losing history.

Decisive counterexample: Dropping a hard requirement to improve the percentage is visible and needs authorization.

Dependency: WP-PROOF-15-1.

WP-PROOF-16-1 — Evidence-bound numeric product grading: contract and first positive path

Bind the existing rubric owner to one frozen scope/profile. Adapt native evidence into typed scored obligations, compute category weights and a literal percentage/letter, and display missing weight and mandatory gates.

Decisive counterexample: A high numeric grade with a failed critical invariant remains BLOCKED.

Dependency: No universal prerequisite; local scope/permissions still required.

WP-PROOF-16-2 — Evidence-bound numeric product grading: negative controls and qualified consumer

Adversarially test grade dilution, stale/wrong-build receipts, missing media, synthetic/live mismatch and failing critical checks. Compare the computed grade with independently inspected outcomes and qualify actual producer/oracle boundaries.

Decisive counterexample: No invented acronym or new registry becomes authority from this amendment.

Dependency: WP-PROOF-16-1.

First integration wave

Choose one owned-game witness (Civis), one permitted gray-box witness (Dino or WorldSphereMod), and one ordinary GUI or CLI consumer. Do not start 24 new bridge implementations. Shared capture/action/evidence interfaces have an owner; engine/platform-specific adapters remain with their products. No product waits for a universal atlas or App Center.

For each witness: define intent and a design fixture; capture one actual user-path action and its state/pixel outcome; seed a known-bad counterpart; bind the receipts to a frozen score rubric; then compare the claimed grade with independently inspected results. A valid transport packet, screenshot filename or scoring-schema pass is not this integration outcome.

A second wave broadens visual classes, design states, reverse lifecycle cases and support profiles. Continue safe independent work while external blockers retain exact owners and wakeup actions. Existing native run and independent assurance requirements remain in force. No fabricated percentages or arbitrary 100% completion labels arise from this package.

All current owner packets

The inherited 24-ID allocation is preserved; no live census refresh was performed for this amendment. Refresh actual source/leases before work. No product grades are assigned here.

Subject	Profile	First proof
.github	publication/defaults	An inherited workflow actually rejects a seeded failure; public links/metadata remain truthful.
AgilePlus	owned GUI/API/CLI	Use host-native action to create/review a scoped work item; restart; reject wrong-root and unauthorized changes.
BytePort	owned desktop/web plus external runtime	Operate one real target through the installed client; verify service state and host pixels plus failed readiness recovery.
CivicWarfare	owned mod plus gray/black host game	User-path onboarding/combat or economy action with visible feedback and save/reload; known missing HUD/asset must fail.
Civis	owned game / white box	Create/populate world; act via player input; verify state and subtle visible feature; save/reopen; negative render and input controls.
Dino	gray-box mod bridge and external capture	Distinctive pack change -> real user action -> stat/combat effect -> visible pixels -> save/reload. An initialization log alone fails.
HeliosCLI	owned fork CLI plus native shell	Execute bounded coding task through actual CLI; cancel/resume; preserve permission boundaries; VHS/transcript evidence.
HeliosLab	owned native/web UI and scoped services	One useful workbench/config task through actual UI with keyboard, error, save/restart and constrained permissions.
HeliosLite	owned coding-agent CLI fork	Real task with stream interruption and non-idempotent-tool boundary; no duplicate action after recovery.
KCode	owned coding-agent CLI fork	Interrupted long task resumes on the intended artifact/workspace without repeating side effects or false completion.
Khostty	owned terminal fork / native graphical	Unicode, resize/reflow, alternate-screen and input sequences on actual native build; tiny glyph clipping and wrong window fail.
KooshaPari	public static/browser surface	A new visitor reaches and installs one current product from the actual page; broken/stale/private link fails.
Melosviz	owned native/web plus media/render providers	Import real media, edit, render/export, reopen and recover; placeholders may qualify pipeline-only scope, never live-render scope.
OmniRoute	owned gateway/web fork	Select provider through actual UI/API, stream/cancel, reconcile quota and error; qualify failover without duplicate billable operations.

Subject	Profile	First proof
PhenoApps	collection of distinct owned applications	Choose one app, capture its complete useful flow and preserve unrelated apps; per-app role-specific grades.
PhenoDesign	shared design tooling/assets	Change one shared design component and inspect at least two real consumers under relevant states.
PhenoFabric	owned multi-device UI/runtime	Invoke a host feature backed by a real peer; verify state/output/recovery, no focus theft and no silent provider swap.
PhenoLab	evaluation/lab CLI or cockpit	Run known-good and known-bad candidates; malformed/missing state or skipped implementation cannot pass.
PhenoRegistry	catalog/source projection	Rename fixture keeps identity; stale/contradictory roles and inaccessible evidence remain explicit.
PhenoShared	package-level capability collection	One shared capability consumed by two products without source copies; pin semantics and negative consumer contracts.
Portage	evaluation framework CLI	Direct upstream/fork matched task with known pass/fail/environment error; imported timestamps are not new evidence.
ShareCLI	supervisor CLI/native GUI	Cancel/restart supervised jobs under foreground load; wrong workspace/no-op and unsafe coalescing fail.
Tracera	owned product graph UI/API	Import a real subject; expose known semantic/visual gap; navigate its evidence; update and restart without fabricated state.
WorldSphereMod	gray-box mod plus black host rendering	Exercise seam/LOD/terrain-color/animation with orbit and movement; small known-bad defect fails even if global similarity stays high.

Source and verification boundaries

This is a targeted source/reference review to refine contracts, not a new all-repository audit. No graphical app, native bridge, game, Steam identity or deployed service was run. The March-dated Dino matrix is historical even though the file was retrieved at a later source revision. An initialized system or enqueued command is weaker evidence than the visible behavior its story promises.

SRC-V14-01 — Dino GameControlCli project

Primary reference: github.com. Exact URL and retrieval scope are in `proof/sources.json`.

Connector search confirms existing named-pipe game-control CLI. Reuse/investigate; no execution here.

SRC-V14-02 — Dino QA validation matrix

Primary reference: github.com. Exact URL and retrieval scope are in `proof/sources.json`.

Fetches source is explicitly last validated March 30, 2026. It distinguishes some visual/runtime failures, but also labels some visible-feature claims programmatic from initialization logs. Treat as historical evidence taxonomy needing refresh, not current bug/grade measurement.

SRC-V14-03 — Civis capture guide and stub

Primary reference: github.com. Exact URL and retrieval scope are in `proof/sources.json`.

Search identifies this path as planned/stub capture helpers; does not imply no other working capture exists. Existing dispatch instructions separately require reading actual PNGs.

SRC-V14-04 — Playwright visual comparisons

Primary reference: playwright.dev. Exact URL and retrieval scope are in `proof/sources.json`.

Official guidance on screenshot baselines and rendering environment variability; use pinned supported versions.

SRC-V14-05 — Playwright screenshots

Primary reference: playwright.dev. Exact URL and retrieval scope are in `proof/sources.json`.

Official full-page, buffer and element capture primitives.

SRC-V14-06 — Bevy externally driven renderer

Primary reference: github.com. Exact URL and retrieval scope are in `proof/sources.json`.

Official example of externally driven offscreen rendering; current main is a research source, not a pinned production dependency.

SRC-V14-07 — Bevy headless renderer

Primary reference: github.com. Exact URL and retrieval scope are in `proof/sources.json`.

Official example includes GPU-image readback and render/main-world frame-latency concerns. No renderer execution here.

SRC-V14-08 — Godot viewport capture

Primary reference: docs.godotengine.org. Exact URL and retrieval scope are in `proof/sources.json`.

Viewport texture capture and waiting for rendering are relevant primitives; verify actual shipped backend.

SRC-V14-09 — Unreal Automation Test Framework

Primary reference: dev.epicgames.com. Exact URL and retrieval scope are in proof/sources.json.

Official framework distinguishes unit/feature/content/screenshot tests and input/functional testing facilities.

SRC-V14-10 — Unreal screenshot comparison

Primary reference: dev.epicgames.com. Exact URL and retrieval scope are in proof/sources.json.

Official settings include tolerances, camera/resolution, noisy-feature handling and diagnostic buffers.

SRC-V14-11 — Unreal automation execution

Primary reference: dev.epicgames.com. Exact URL and retrieval scope are in proof/sources.json.

Official client/editor execution and report export facilities; editor success is not automatically packaged-client proof.

SRC-V14-12 — Unity Test Runner

Primary reference: docs.unity3d.com. Exact URL and retrieval scope are in proof/sources.json.

Historical official documentation establishes Edit/Play/target-player distinction. It is not a current package-version recommendation. Inspect installed package docs before command selection.

SRC-V14-13 — W3C accessibility evaluation tools

Primary reference: www.w3.org. Exact URL and retrieval scope are in proof/sources.json.

Tools assist evaluation but cannot determine all accessibility aspects; qualified judgment remains needed for some criteria.

SRC-V14-14 — Supplied v1.3 archive

Source: ../references/v1.3-input-identity.json

Previous package is preserved externally and identified by digest in this amendment; prior policies and 24 owner folders are retained, not a live refreshed census.

All numerical examples added in v1.4 are synthetic. Grading weights/letters are proposed policy, not empirical industry standards. Source platform features do not establish that any Phenotype adapter correctly uses them.